

RESEARCH ARTICLE

Density dependence modulates climate change impacts on eastern Bering Sea snow crab

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Handling Editor: Maria Paniw**Abstract**

1. Populations change as a result of many interacting processes, each of which may be moderated by density dependence and/or climate. Models and projections of population dynamics support conservation and economic planning, but the extent to which the influence of climate or density is incorporated varies.
2. We develop a population dynamics model for eastern Bering Sea snow crab (*Chionoecetes opilio*) that fits to data collected from 1982 to 2023 to estimate variability in recruitment and mortality. We then use generalized additive models (GAMs) to explain the variability in recruitment, mortality and maturity using environmental covariates and density. Finally, we project the population forward under various assumptions about climate and density dependence on recruitment, mortality and maturity.
3. Feedbacks among population processes result in different short- and long-term population trajectories. For example, including density dependence in mortality projections allows for a stronger rebound in the population in the short term, but the harmful effects of warming outweigh the protective effect of lower density in the long term.
4. Projected long-term crab abundances were much lower than historically observed under receding ice cover in the Bering Sea regardless of the combination of processes driven by density and climate.
5. *Synthesis and applications.* Our results underscore the need for management capable of addressing the impacts of both density and climate effects on multiple processes in harvested populations. Non-stationarity in drivers of population dynamics (e.g. climate change) will further affect the ability of management to achieve sustainable yields. Current frameworks built around equilibrium-based maximum sustainable yield are not well-suited to address either of these issues.

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KEYWORDS

Bering Sea snow crab, climate change, fisheries management, population collapse, population dynamics model, stock assessment, time-varying mortality

1 | INTRODUCTION

Populations grow and shrink over time due to the net effect of many biological processes. Reproductive processes determine recruitment, or the number of young individuals joining the population. Growth and maturity processes determine the size of the animals in the population and their contribution to reproduction. Predation, disease and metabolic needs contribute to mortality, which determines the number of individuals exiting the population. Each of these processes can be influenced by external climate factors (e.g. sea surface temperature or rainfall) or internal factors (e.g. density dependence). The feedbacks among these processes can provide both stabilizing (e.g. density-dependent growth, Lorenzen & Enberg, 2002; Stenevik et al., 2022; Matthias et al., 2018) and destabilizing effects (e.g. Allee effects that destabilize dynamics at low population densities; Stephens et al., 1999; Berec et al., 2007).

Climate change and human impacts are altering these population processes. Recruitment of marine species has been linked to a range of climate variables (e.g. sea surface temperature; Britten et al., 2015; Le Bris et al., 2018), large-scale indices of climatic variation (Faillettaz et al., 2019), plankton abundance and composition (Beaugrand et al., 2003) and ice extent (Szuwalski et al., 2021). North Sea Atlantic cod (*Gadus morhua*) growth rates and size-at-age have been linked to changes in temperature (Neuheimer & Grønkjær, 2012), while Bering Sea groundfish growth rates show strong temporal variation related to environmental change (Stawitz et al., 2015). Changes in maturity have been related to density-dependent effects for silver hake (*Merluccius bilinearis*; Helser & Almeida, 1997), exploitation in herring (*Clupea harengus*; Engelhard & Heino, 2004) and environmental drivers such as temperature and prey-availability for fresh water perch (*Perca flavescens*; Kaemingk et al., 2014). Changes in mortality have been linked to heatwaves (e.g. snow crab; Szuwalski, Aydin, et al., 2023) and blue petrel (*Halobaena caerulea*; Barbraud & Weimerskirch, 2003). Increased hurricane activity has been linked to increased mortality in dolphins in the Bahamas (Coxon et al., 2022).

Projections of populations under climate change can support conservation and economic planning. Knowledge of plausible futures of the populations under management can be used to anticipate and adjust for potential problems such as population collapse or increase. Population projections often use estimated parameters from models fit to historical data to project a population forward (Hollowed et al., 2009; Holsman et al., 2020; Punt et al., 2014). Some projections include climate-linked parameters and use global climate models to inform future variation in that process (A'mar et al., 2009; Szuwalski & Punt, 2013). However, most commonly only a single process (e.g. recruitment or growth) is linked to climate and/

or population density in projections of harvested marine resources (Punt et al., 2014), in spite of the possibility that multiple processes may be affected. This approach is somewhat different from in terrestrial ecology where the exploration of the interaction between density dependence and climate drivers of unexploited populations is more common (e.g. Ickin et al., 2023; Paniw et al., 2018).

We demonstrate how considering density dependence and climate drivers in multiple processes can change perceptions about the future of a harvested population using eastern Bering Sea snow crab (*Chionoecetes opilio*) as a case study. Snow crab is a useful motivating example because the fishery for it closed for the first time ever in 2021 after a marine heatwave precipitated a collapse (Szuwalski, Aydin, et al., 2023). This collapse led to the need for a rebuilding plan that requires population projections under different climate scenarios and fishing strategies.

2 | MATERIALS AND METHODS

2.1 | Overview

The snow crab is a stenothermic species, and juveniles in particular are associated with bottom waters colder than 2 degrees Celsius (Dionne et al., 2003). The eastern Bering Sea population is widely distributed across the shelf and is one of the most abundant animals captured in the NMFS bottom trawl surveys (Figure 1; Zacher et al., 2024). Snow crab have a maximum age of ~20 and are strongly sexually dimorphic, with males growing to almost twice the size of females (Szuwalski, 2023). A pot fishery occurs during the winter in which only the largest male individuals are retained. A fishery has occurred for snow crab in the eastern Bering Sea since the 1970s, but production substantially increased during the 1990s (Figure 1).

We used an integrated population dynamics model to estimate variation in mortality (partitioned into 'other' and 'fishing') and recruitment over time for snow crab in the eastern Bering Sea. Other mortality accounts for processes such as predation, disease, non-directed fishery bycatch, and cannibalism. The population dynamics model was fitted to both survey and fishery data (see Supporting Information for a full description of the data). We then used generalized additive models (GAMs; Wood, 2011) to relate estimated variation in 'other' mortality and recruitment to environmental covariates and density. GAMs were also used to relate variation in the observed probability of maturing to environmental covariates and density. The population was then projected under two climate scenarios and four scenarios for variability in population processes, all while being fished at the average estimated fishing mortality over the study period.

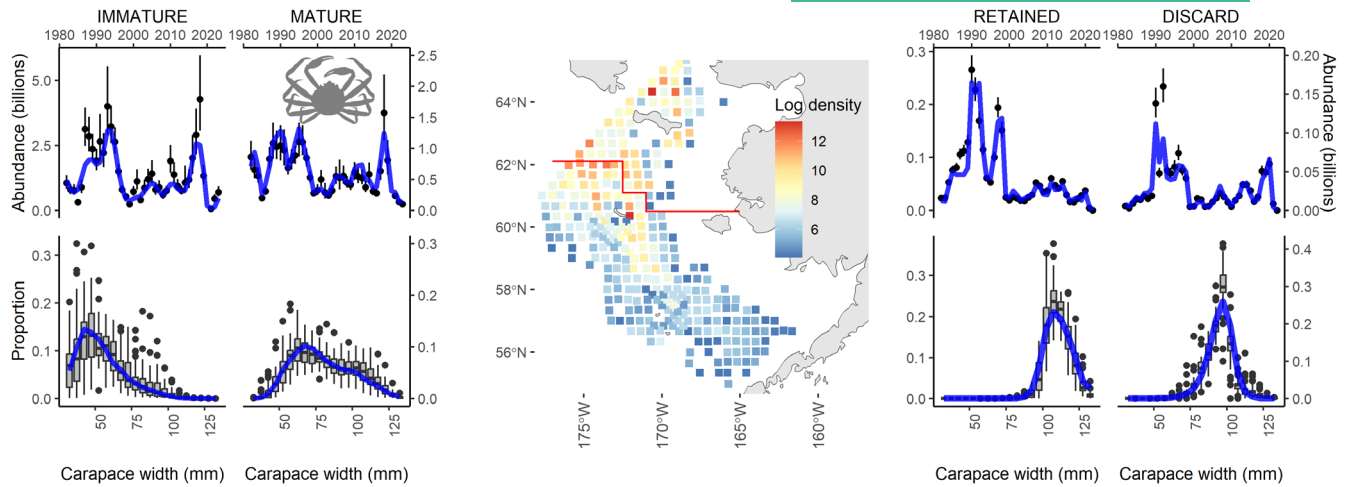


FIGURE 1 Data from the survey and fishery with model fits. Left panel are the immature and mature indices of abundance and size composition data from the survey. Middle panel is the observed density of snow crab in the eastern Bering Sea during the 2019 survey; each square is a survey station and the colour is the log density in numbers per km². Right panel are the retained and discarded numbers and size-composition data in the directed fishery for snow crab. Observations are black dots with 95% confidence intervals for abundances and box plots for size-composition data, which are aggregated over time. Blue lines represent the model fits to the data.

2.2 | Population dynamics model

Our model was fit to abundance and size-composition data for immature and mature male snow crab collected in National Marine Fisheries Service (NMFS) bottom trawl surveys (1982–2023). Retained and discarded numbers-at-size of snow crab from this fishery were used to account for fishery removals. In addition to estimating time-varying ‘other’ mortality (separately for immature and mature crab) and recruitment within the model, parameters associated with several other processes were also estimated, including fishing mortality, selectivity parameters for the survey and fishery, and the proportion of recruitment entering the first size bins.

The population dynamics model used here is based closely on that used in the stock assessment (Szuwalski, 2023). The model tracks numbers of male crab at size s by maturity state m over time t ($N_{t,s,m}$) with 5 mm size bins. The model is divided into two ‘periods’: pre-fishery and post-fishery. The timing of the fishery is denoted by the month in which it occurs (q below). Other mortality is the only population process that occurs in the first 7 months of a given year (the crab year starts in July when the NMFS bottom trawl survey takes place):

$$n_{t=y+\frac{q}{12},s,m} = N_{t=y,s,m} e^{-\frac{q}{12}M_m} \quad (1)$$

Fishing is assumed to occur as a pulse fishery in which the crab captured and brought on deck of fishing vessels, C_{cap} , are a function of capture selectivity, S_{cap} , the number of crab in the Bering Sea at the time of fishing, and the fishing mortality applied, F_t . A retention ogive, S_{ret} , is applied to the captured crab to determine what fraction of crab at a given size are retained, $C_{ret,y}$, and what fraction are discarded back into the ocean. A discard mortality, d_{mort} , of 0.3 is applied to the crab returned to the ocean.

$$C_{cap,y,m} = n_{t=y+\frac{q}{12},s,m} (1 - e^{-F_t S_{cap,s}}) \quad (2)$$

$$C_{ret,y,m} = C_{cap,y,m} S_{ret,s} \quad (3)$$

$$n_{t=y+\frac{q}{12},s,m} = n_{t=y+\frac{q}{12},s,m} e^{-F_t S_{cap,s}} + (1 - d_{mort}) C_{cap,y,m} (1 - S_{ret,s}) \quad (4)$$

Capture selectivity and retention selectivity are logistic functions of size in which the slope and size at which 50% selection are estimated.

$$S_{cap,s} = \frac{1}{1 + e^{-S_{slope,cap}(size_s - S_{50,cap})}} \quad (5)$$

$$S_{ret,s} = \frac{1}{1 + e^{-S_{slope,ret}(size_s - S_{50,ret})}} \quad (6)$$

Growth occurs after the fishery and is represented in the model by multiplying the vector of moulting crab at size s by a size-transition matrix, $X_{s,s'}$, that specifies the size to which crab grow given an initial size. The size transition matrix $X_{s,s'}$ used here was developed outside of the model using the growth increment data used in the stock assessment and is very similar to the one estimated in the assessment (Szuwalski, 2023). Snow crab has a terminal moult to maturity, after which they do not grow again (Tamone et al., 2005). The probability of maturing (ρ) and moulting probability (z) are specified within the model based on observations from the survey using methodology from Richar and Foy (2022). All immature individuals are assumed to moult (i.e. $z = 1$). Growth results in two temporary vectors of numbers at size where immature crab are denoted by the subscript $m = 1$; mature crab are denoted by the subscript $m = 2$:

$$n_{t=y+\frac{q}{12},s,m=1} = z(1 - \rho_s) X_{s,s'} n_{t=y+\frac{q}{12},s,m=1} \quad (7)$$

$$n_{t=y+\frac{q}{12},s,m=2} = z\rho_s X_{s,s'} n_{t=y+\frac{q}{12},s,m=1} + n_{t=y+\frac{q}{12},s,m=2} \quad (8)$$

Recruitment is a freely estimated vector composed of a mean and deviations around that mean. Recruitment enters the model in the first three size bins with relative estimated proportions allocated to a given size bin ($\theta_{1,2,3}$):

$$n_{t=y+\frac{a}{12},s=1,2,3,m=1} = n_{t=y+\frac{a}{12},s=1,2,3,m=1} + r_y \theta_{1,2,3} \quad (9)$$

Finally, the remaining portion of the year's other mortality is applied to the population after growth, moulting, recruitment and fishing occur. Note that this can allow a crab to experience two different levels of mortality within a given year as it undergoes terminal moult.

$$N_{t=y+1,s,m} = n_{t=y+\frac{a}{12},s,m} e^{-\frac{12-a}{12} M_{t,s,m}} \quad (10)$$

Estimated survey selectivity (i.e. the proportion of crab captured at size by the survey trawls) is applied to the model predictions of numbers at size to compare to observations during optimization. The likelihoods, penalties, and priors used in this model can be seen in the [Supporting Information](#) along with data descriptions, simulation testing, model diagnostics and further details on the modelling of mortality, selectivity, growth and recruitment. The ADMB model that uses the above equations to fit to the observed data can be found in the github repository noted at the end of this document.

2.3 | Modelling variation in population processes

We used generalized additive models in the R programming language (GAMs; package *mgcv*; Wood, 2011) to explain the variability in other mortality, recruitment and the probability of maturing using climate variables and estimates of crab density. Previous studies examining snow crab recruitment identified a positive relationship with ice extent and a negative relationship with the Arctic Oscillation (a measure of the volatility in the jet stream; Szuwalski et al., 2021). Szuwalski, Aydin, et al. (2023) identified positive relationships between the mortality of snow crab and bottom temperature and crab density. The probability of maturing has been linked to the density of large males in eastern Canadian populations (Mullowney & Baker, 2021). Based on these studies, we tested for relationships between our estimates of mortality, recruitment, and the probability of maturing and temperature-related variables (ice extent and cold pool), the Arctic Oscillation (AO) and our model estimates of population density. Models were first fitted in which all potentially relevant covariates were included in a model of the form:

$$p_{x,y} = s(\theta_{x,y}) + \varepsilon_{x,y}$$

where $p_{x,y}$ is a process to be explained (i.e. recruitment, maturity, or mortality), $s()$ is a smoothing function based on thin-plate splines, $\theta_{x,y}$ is a matrix of environmental covariates (each of which have a separate smooth estimated), and $\varepsilon_{x,y}$ is unexplained error. Then relevant covariates for each process were included in a factorial design and models were compared via Akaike's information criterion (AIC, Akaike, 1974). The families used as link functions for each process were different: a Tweedie was used for mortality, a negative binomial was used for

recruitment, and a beta was used for maturity. See the [Supporting Information](#) for further discussion about diagnostics and covariate construction.

2.4 | Projecting under climate change scenarios

Finally, we projected the snow crab population forward 40 years beginning from the final year of the model under two potential climate scenarios: one in which environmental conditions were sampled from historical conditions and one in which sea ice in the Bering Sea declined in a manner similar to projections from global climate models under SSP585 (Figure S1). The scenarios represent a future in which climate change mitigation is successful and future climate is similar to the past and a future in which climate change is not successfully mitigated and future climate does not resemble the past. Projections were repeated 100 times and fishing mortality rates were set to the estimated average over the study period (see [Supporting Information](#), for further details). Four combinations of variability in population processes driven by environment and density effects from the best-fitting GAMs were considered: variable recruitment (Rec), variable recruitment and mortality (Rec + M), variable recruitment and maturity (Rec + Maturity), and variable recruitment, maturity and mortality (Rec + Maturity + M). The projected abundance of mature males, other mortality, probability of maturing and recruitment were recorded for comparison across scenarios.

3 | RESULTS

The population dynamics model fitted both the survey and catch data well, which is perhaps unsurprising given the estimation of time-varying mortality and recruitment (Figure 1). Estimated other mortality for mature crab was higher on average and more variable than for immature crab (Figure 2; see [Supporting Information](#), for discussion). Estimated recruitment was punctuated by two large pseudocohorts (i.e. groups of similarly sized crab; Figure 2).

GAMs including climate variables and density explained the variability in mortality, recruitment and maturity better than those without based on AIC (Figure 2). The best fitting GAMs for mortality used density and lagged ice cover to explain 59% and 27% of the deviance for immature and mature mortality, respectively. Less ice and increases in density were associated with higher mortality for both mature and immature crab (Figure 3). GAMs using the Arctic Oscillation, ice cover and density of mature crab fit the estimated recruitment best and explained 48% of the deviance. Negative phases of the AO and more ice were associated with higher recruitment (Figure 3). GAMs using carapace width, density of small mature males, density of large mature males and ice cover fit the probability of maturing data best and explained 94% of the deviance. Colder temperatures and higher densities of small and large crab were associated with lower probabilities of undergoing terminal moult (Figure 3).

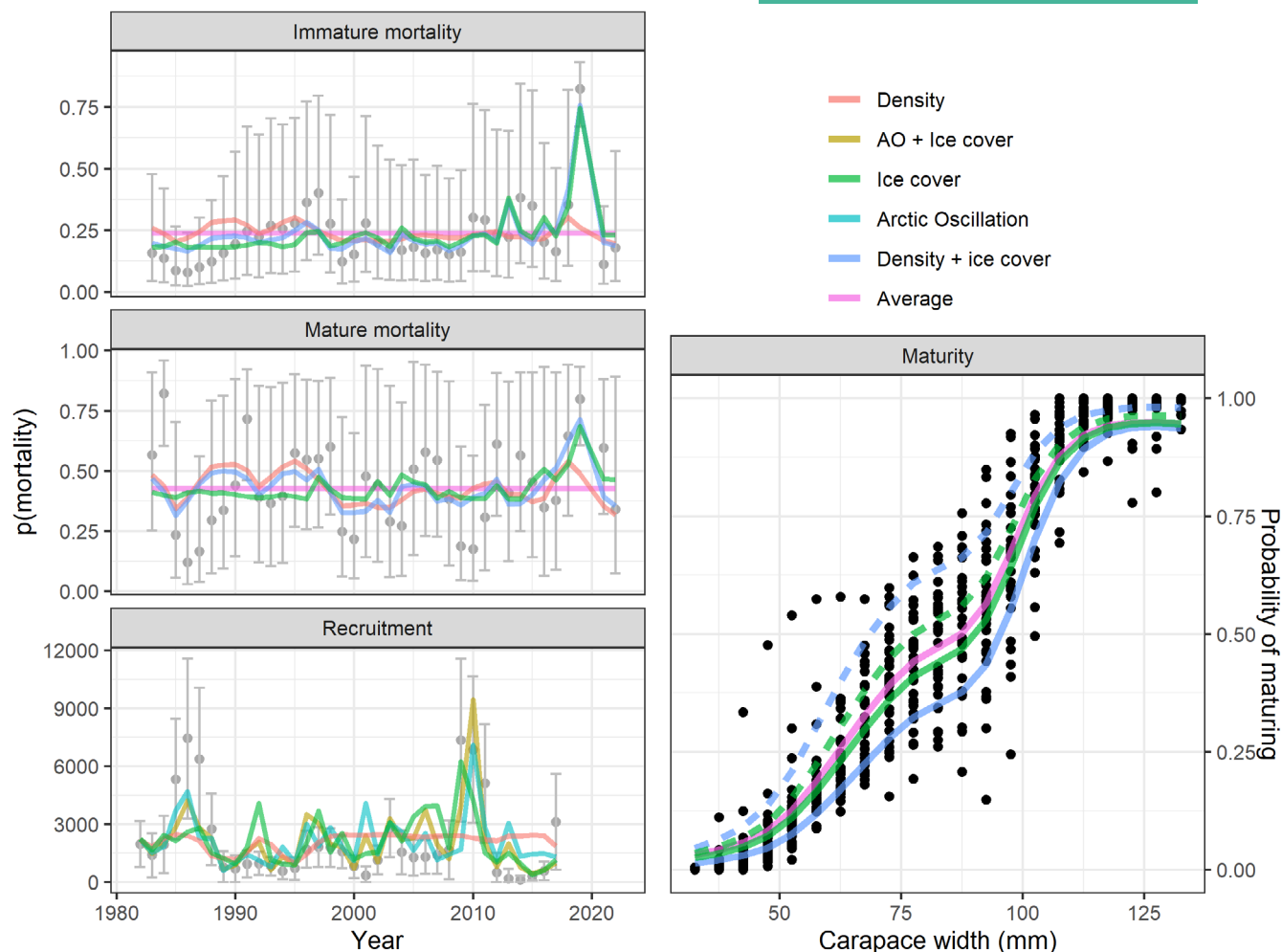


FIGURE 2 Density and climate effects in population processes. Black dots and/or lines are estimated population processes from the population dynamics model (recruitment and mortality) or observations from the NMFS survey (probability of maturing). Coloured lines are the fits from GAMs incorporating a given set of covariates. Coloured lines for the probability of maturing in which environmental covariates are included represent the probability given the combination of variables that produce the highest and lowest observed values (solid and dashed, respectively). Units for recruits are 1,000,000 individuals, other mortality is a yearly rate, and maturity is the probability of having matured at 75 mm carapace width.

Within our projections, the median trajectories of the number of mature males in the population and other population processes were similar to historical dynamics when environmental conditions were sampled from historical values (Figure S2). Ensuring this is the case is an important check on the model before projecting under directional change in climate drivers. When ice declined over the projection period, the stock declined after a brief rebound. The magnitude of this rebound was smaller when only recruitment was density-dependent and driven by climate (Figure 4). Declines in recruitment related to receding ice drove the decline of the stock in projections, and increases in mortality related to disappearing ice compounded the decline after ~2040 in scenarios in which other mortality was linked to climate and density. The probability of maturing at size over the projection period increased when recruitment and maturity were linked to climate and density, but declined somewhat in the 2040s if mortality was also linked to climate and density. Dramatically lower abundances under receding ice would affect the fishery given potential fishery closures.

4 | DISCUSSION

Both density and climate variables were important covariates for population processes that determine the dynamics of snow crab. Feedbacks in our models were determined by the directionality and strength of these relationships to climate and density. Mortality increased with density, recruitment was dome-shaped with respect to density, and the probability of maturing depended on both the numbers of small and large mature crab. Low projected population density would suggest that mortality should remain low in the absence of climate effects. However, when accounting for increasing temperatures, mortality increased towards the end of the projections even while abundance remained low. The opposite directions of the relationships to mortality between density and temperature resulted in a stable mortality rate for more of the projection than if only one driver was considered and that rate was lower than the median of the historical estimates. The probability of maturing also

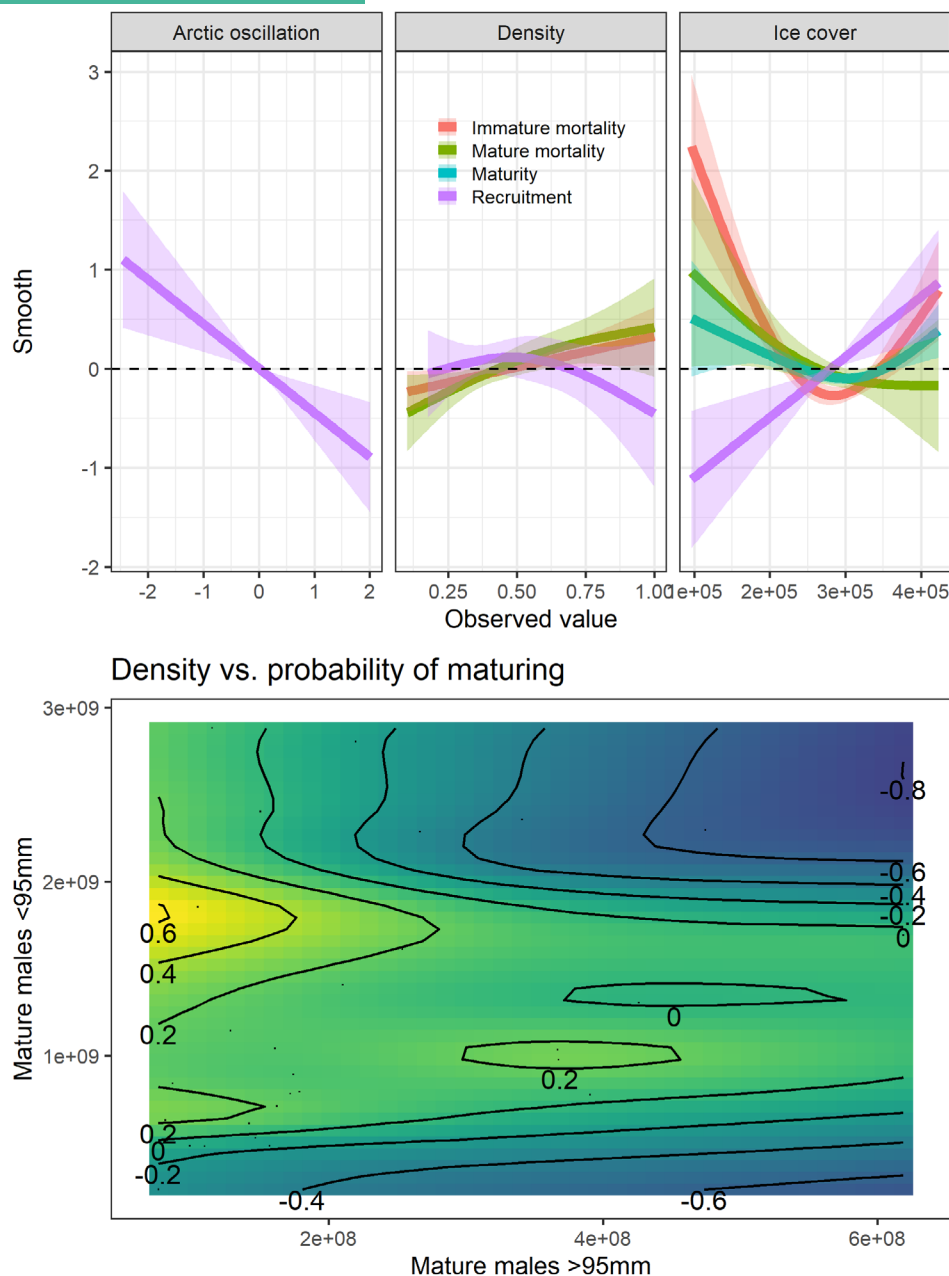


FIGURE 3 Shapes of estimated smooths relating population process to covariates. In the top row, each panel represents a covariate, with each colour corresponds to a population process (shading represents one standard error). The bottom plot shows the surface relating the probability of maturing to the density of large and small males. Lighter yellow indicates a higher probability of maturing; darker blue indicates a lower probability.

exhibited several feedbacks: it was related to the number of small and large mature males, which was in turn influenced by mortality. Linking mortality to density and the environment resulted in an increase in the number of mature males in the short term, which was then related to an increase in the probability of maturing.

Global climate models and current carbon mitigation efforts suggest an ice-free future in the eastern Bering Sea (Cheng et al., 2021; Crawford et al., 2021), which bodes poorly for the long-term future of snow crab in this region. However, our projected short-term prospects for snow crab changed depending on which processes were assumed to vary with the climate and density, and

the difference in short-term projections among scenarios could have implications for management decisions. In the United States, a rebuilding plan is legally required to return the population to a target biomass when a managed population declines beneath a specified biomass (Magnuson-Stevens Fishery Conservation and Management Reauthorization Act of 2006, 2007). The decisions required around rebuilding plans can be different when a stock is deemed unable to return to that biomass even in the absence of harvest when compared to stocks that have the potential to rebuild, so identifying if a stock is able to rebuild is important to management.

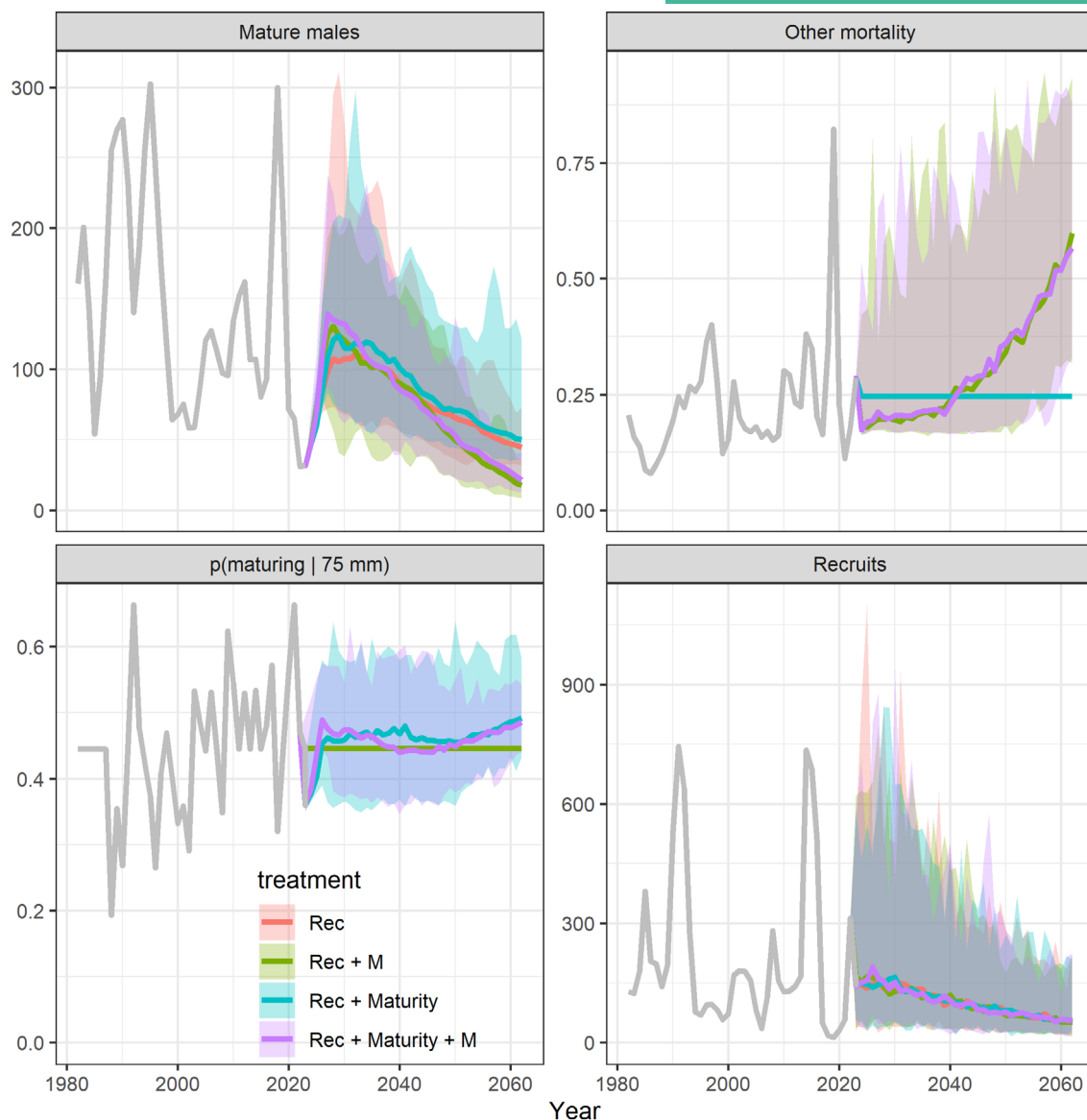


FIGURE 4 Projections of population dynamics under declining ice cover. Grey lines are the historical maximum likelihood estimates (or observations in the case of maturity). Scenarios (coloured lines) are distinguished by which biological processes are allowed to vary based on the best-fitting GAM described in the text. Solid lines are the medians over 100 simulations; shaded areas represent the inter-simulation variability in each quantity for a given scenario. Units for recruits and mature males are 10,000,000 individuals, other mortality is a yearly rate, and maturity is the probability of having matured at 75 mm carapace width.

Decisive management actions based on projections like ours can be difficult due to uncertainty associated with the correlative relationships identified between the environment and population processes changing over time. For example, recruitment processes and their relationship to the environment have a long history of study in fisheries, yet recruitment–environment relationships are infamously unstable. Myers (1998) found that only 28 of 74 significant recruitment–environment relationships reported in the literature retained their significance when new data were added to the time series. Uncertainties are compounded when more than one process is linked to the environment and density, not only because there are more relationships that could break down, but also because changing the relative strengths of

the climate drivers and density dependence on the processes alters feedback structures.

Even if these uncertainties are ignored, non-stationary projections of populations can be difficult to incorporate into management systems designed for equilibrium assumptions. Maximum sustainable yield is the prevailing management paradigm in fisheries science and many management plans have provisions to consider the current climate environmental conditions in setting management targets (e.g. Common Fisheries Policy in the European Union [Lado, 2016] and the Magnuson-Stevens Act in the USA [MSA, 2007]). Under climate change-driven changes in productivity, adapting management targets to incorporate current climate conditions would have the effect of increasing anthropogenic

pressure on many populations experiencing climate stress compared to using climate-naïve targets (Bessel-Brown et al., 2022; Szuwalski, Aydin, et al., 2023; Szuwalski, Hollowed, et al., 2023). This seems like a counter-intuitive management response, yet it can be a consequence of attempting to apply equilibrium-based management targets in a non-stationary environment (Szuwalski & Hollowed, 2016).

In addition to difficulties identifying appropriate management targets, projections of continued decline of a population raise difficult management questions for which science-based answers are scarce. If we place confidence in an abrupt decline of snow crab sometime in the next 20 years (or even 2 years), how should that influence management today? One possible management response is maintaining fishery closures to allow the population every chance to adapt and survive in new climates. Another possible management response is harvesting the commercially viable biomass that would otherwise die as a result of inhospitable environmental conditions. Our current management frameworks do not have quantitative methods to address these questions, but they are needed. One of the key uncertainties in these decisions is the adaptive capacity of species experiencing climate stress.

Attempting to predict the future trajectories of populations given observed relationships between population processes and the environment is challenging, but ignoring potential climate drivers can have even more undesirable consequences. Unrealistic expectations of population futures can confound conservation planning, resulting in chasing unreachable management targets towards which a lack of progress may be seen as a failure of management. Ignoring climate drivers can result in economic plans that are not sufficiently flexible to respond to the coming changes in ecosystems (e.g. capitalization in fishing fleets designed to capture species that will no longer be present in the ecosystem). One of the difficulties in managing populations responding to climate stress is that resource managers do not possess a lever for the fundamental process driving change in their systems: atmospheric CO₂ (Mullowney et al., 2020). The lack of a clear picture of what to expect for future mitigation confounds efforts of management and clarity does not appear forthcoming given uncertain and unfulfilled promises by governments of key CO₂ producing nations (Brown et al., 2019).

Despite the policy challenges associated with managing climate-threatened populations, longitudinal data collection targeted at identifying relationships such as those described here for snow crab can help inform plausible future population scenarios. The continued collection of these data should be a key priority in institutions charged with tracking climate-driven changes in population dynamics. Data collection at a wider spatial scale than the population currently resides is particularly important so that distributional shifts (or lack of shifts) can be captured. Analysis of population dynamics at range edges and along spatial gradients may be useful for understanding population responses to novel environmental conditions (Fredston et al., 2021). Populations may be more likely to approach extreme abundances (both high and low) that have been rarely observed in the past under the coming changes in climate.

Allee effects may become more prevalent and density dependence may become stronger under less hospitable environmental conditions (Berec et al., 2007). Some processes are easier to study because they can be directly observed (e.g. growth and recruitment), while other processes (e.g. mortality and selectivity/catchability) can be more difficult because they are confounded and must be inferred from commonly available data rather than directly observed (Szuwalski, 2022). More focussed data collection may also be required to identify changes in processes that are difficult to characterize using current data sources. Estimating annual mortalities, for example, could be improved by implementing large-scale mark-recapture studies (Queroue et al., 2021) and other important (and potentially changing) life-history information such as maturity could be collected alongside these efforts.

Our example here is a harvested marine population, but the importance of considering density dependence and climate forcing of population processes is applicable to all populations experiencing the effects of climate change. Identifying three processes related to both climate and density dependence as we have done here for snow crab is unusual, but these sorts of relationships may be easier to identify as species become subject to environmental conditions that are outside of historical ranges (Berec et al., 2007). It is important to note that density dependence will not universally stabilize mortality in the short term as occurred here; population-level outcomes depend on the direction and relative strengths of processes. For example, Baltic blue mussels (*Mytilus edulis*) displayed the opposite outcome in which density dependence appears to exacerbate the harmful effects of climate change (Jaatinen et al., 2021).

Identifying climate drivers of populations and considering them in projections of populations will be important as climate change becomes more pronounced (Hollowed et al., 2020). The identification of these drivers can be difficult to incorporate into currently used management systems that rely on equilibrium assumptions, but examples of complex intrinsic and extrinsic drivers are needed as new management methodologies are developed. Feedback between climate drivers and density dependence could play an important role in the dynamics of populations under historically unobserved conditions. Considering these drivers and feedbacks may help us understand the future trajectory of a population better, but this understanding is only the first step. Models and projections such as those presented here will likely be a centre point for the cooperative, transparent and adaptive decision-making that will be needed to address the complex natural resource problems on the horizon. However, these projections do not provide answers to the question of 'what do we do?' in the same way that equilibrium frameworks based on maximum sustainable yield did before climate change began strongly impacting harvested populations.

AUTHOR CONTRIBUTIONS

Cody Szuwalski conceived the ideas and designed the methodology. Cody Szuwalski, André E. Punt, Jon Richar, Laurinne J. Balstad, Maxime Olmos, Baptiste Aglave, William Stockhausen and Matthieu Veron wrote code that was either referenced or used in the analyses.

Cody Szuwalski led the data analysis and manuscript preparation. All authors contributed critically to the drafts and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The code and data used to produce this manuscript can be found in Zenodo at: <https://doi.org/10.5281/zenodo.18475044> (Szuwalski et al., 2026).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. AIC by model by process (columns) and covariates (rows).

Figure S1. Historical (blue) and projected future ice scenarios in the Bering Sea (red and green). Solid lines are the median condition over simulations; shading are the inter-simulation ranges.

Figure S2. Projections of population dynamics under resampled historical environmental conditions.

Figure S3. Observed probability of maturing (blue lines) in a given year.

Figure S4. Observed relationship between pre- and post-molt size for male crab (left; dots are observations of individual crab, line is a linear regression) and the resulting size transition matrix used in the population dynamics model (right).

Figure S5. Map showing the survey location and timing of BSFRF experimental survey data collection.

Figure S6. Inferred selectivity at size by year from the BSFRF selectivity experiments. Black line and grey shading are a GAM fit to the data.

Figure S7. Estimated survey selectivity and fishery selectivity.

Figure S8. Bycatch of snow crab in non-directed Bering Sea fisheries.

Figure S9. Energetic condition of snow crab based on water content in the hepatopancreas reported in percent dry weight (reproduced from Fedewa et al., 2024).

Figure S10. Estimated mortalities when auxiliary information about the timing of the collapse is not used to tune the model.

Figure S11. Estimated indices of abundance and size composition data when auxiliary information about the timing of the collapse is not used to tune the model.

Figure S12. Model fits to individual years of size composition data for immature males captured in the survey.

Figure S13. Model fits to individual years of size composition data for mature males captured in the survey.

Figure S14. Model fits to individual years of size composition data for retained crab in the directed fishery.

Figure S15. Model fits to individual years of size composition data for discarded crab in the directed fishery.

Figure S16. Estimated recruitment (black lines) vs. simulated (red line).

Figure S17. Estimated fishing mortality (black lines) vs. simulated (red line).

Figure S18. Estimated immature maturity (black lines) vs. simulated (red line).

Figure S19. Estimated mature maturity (black lines) vs. simulated (red line).

Figure S20. Fits to the survey indices of abundance and size composition data when the mean average other mortality is not estimated.

Figure S21. Diagnostics for best GAM fit to estimated recruitment.

Figure S22. Diagnostics for best GAM fit to estimated immature mortality.

Figure S23. Diagnostics for best GAM fit to estimated mature mortality.

Figure S24. Diagnostics for best GAM fit to the observed probability of maturing.

Figure S25. A comparison of estimated time series of immature and mature mortality between this study and previous studies and the relationships between density, temperature, and those estimates.

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